

10)(i) Find the values for  $k$  which equations.

$$kx + y + z = 1$$

$$x + ky + z = 1$$

$$x + y + kz = 1$$

a) infinitely many solution.

b) no solution.

c) unique solution.

$$[A|B] = \left[ \begin{array}{ccc|c} k & 1 & 1 & 1 \\ 1 & k & 1 & 1 \\ 1 & 1 & k & 1 \end{array} \right]$$

$$\sim \left[ \begin{array}{ccc|c} 1 & k & 1 & 1 \\ k & 1 & 1 & 1 \\ 1 & 1 & k & 1 \end{array} \right] \quad R_1 \leftrightarrow R_2$$

$$\sim \left[ \begin{array}{ccc|c} 1 & k & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ 0 & 1-k & k-1 & 0 \end{array} \right] \quad \begin{array}{l} R_2 \leftrightarrow R_2 - kR_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[ \begin{array}{ccc|c} 1 & k & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ 0 & 0 & k^2+k-2 & k-1 \end{array} \right] \quad R_3 \rightarrow (1+k)R_3 - R_2$$

$$\sim \left[ \begin{array}{ccc|c} 1 & k & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ 0 & 0 & \frac{(k+2)}{(k-1)} & k-1 \end{array} \right]$$

It is in Echelon Form.

a) If  $k=1$ , then

$$R(A) = R[A, B] = 1 < n = 3$$

$\therefore$  The system of Egn has infinite no. of sol.

b) If  $k=-2$ , then

$$R[A] = 2 \neq R[A, B] = 3$$

$\therefore$  The system of Egn has no solutions.

c) If  $k \neq 1, k \neq -2$  then

$$R[A] = R[A, B] = n = 3$$

$\therefore$  The SOE has unique solutions.

10) (ii) For what values of  $\lambda, \mu$  does SOE

$$x + y + 2z = 2$$

$$2x - y + 3z = 2$$

$$5x - y + \lambda z = \mu \text{ have}$$

a) IM S.

b) NS.

c) US.

$$[A|B] = \left[ \begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 2 & -1 & 3 & 2 \\ 5 & -1 & \lambda & \mu \end{array} \right]$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 0 & -3 & -1 & -2 \\ 0 & -6 & \lambda-10 & \mu-10 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 5R_1 \end{array}$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 0 & -3 & -1 & -2 \\ 0 & 0 & \lambda-8 & \mu-6 \end{array} \right] R_3 \rightarrow R_3 - 2R_2$$

It is in Echelon Form.

(i) If  $\lambda=8, \mu=6$ , then

$$R[A] = R[A, B] = 2 < n = 3$$

$\therefore$  The SDE has IMS.

(ii) If  $\lambda=8, \mu \neq 6$ , then

$$R[A] = 2 \neq R[A, B] = 3$$

$\therefore$  The SDE has NS.

(iii) If  $\lambda \neq 8, \mu \neq 6$ , then

$$R[A] = R[A, B] = n = 3$$

$\therefore$  The SDE has US.

HP. 10)(iii) Find the values of  $k$  for which the equations.

$$kx - 2y + z = 1$$

$$x - 2ky + z = -2$$

$$x - 2y + kz = 1 \quad \text{have}$$

a) IMS.

b) NS.

c) US.

iv) For what values  $\lambda, \mu$  does SOE expression

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu \text{ have same.}$$

a) IMS

b) NS

c) US

$$[A|B] = \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array} \right]$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array} \right] R_3 \rightarrow R_3 - R_2$$

It is in Echelon Form.

a) If  $\lambda = 3, \mu = 10$ , then

$$R[A] = R[A, B] = 2 < n = 3$$

$\therefore$  The system has infinite no. of solution.

b) If  $\lambda = 3, \mu \neq 10$ , then

$$R[A] = 2 \neq R[A, B] = 3$$

$\therefore$  The system has no solution.

(iii) If  $\lambda \neq 2$ ,  $\mu = \text{any value} \in \mathbb{R}$

$$\therefore R[A] = R[A, B] = 3 = n$$

$\therefore$  The System has unique solution.

11) Find the non-trivial solution for following system of Equations. if it is exist.

$$(i) \quad x - y + 2z - 3w = 0$$

$$3x + 2y - 4z + w = 0$$

$$5x - 3y + 2z + 6w = 0$$

$$6x - 9y + 14z - 2w = 0$$

$$[A|B] = \left[ \begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 3 & 2 & -4 & 1 & 0 \\ 5 & -3 & 2 & 6 & 0 \\ 6 & -9 & 14 & -2 & 0 \end{array} \right]$$

$$\sim \left[ \begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 0 & 5 & -10 & 10 & 0 \\ 0 & 2 & -8 & 21 & 0 \\ 0 & -8 & 12 & 1 & 0 \end{array} \right] \quad \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 5R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\sim \left[ \begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 0 & 1 & -2 & 2 & 0 \\ 0 & 2 & -8 & 21 & 0 \\ 0 & -8 & 12 & 1 & 0 \end{array} \right] \quad R_2 \rightarrow R_2 / 5$$

$$\sim \left[ \begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 0 & 1 & -2 & 2 & 0 \\ 0 & 0 & -4 & 17 & 0 \\ 0 & 0 & -4 & 17 & 0 \end{array} \right] \quad \begin{array}{l} R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 + 8R_2 \end{array}$$

$$\sim \left[ \begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 0 & 1 & -2 & 2 & 0 \\ 0 & 0 & -4 & 17 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad R_4 \rightarrow R_4 - R_3$$

It is in Echelon Form.

$|A| = 0$ , expanding along  $R_4$ .

$\therefore$  The system has non trivial solution.

$$(A) \Rightarrow R_1 \Rightarrow x - y + 2z - 3w = 0 \quad \text{--- (1)}$$

$$R_2 \Rightarrow y - 2z + 2w = 0 \quad \text{--- (2)}$$

$$R_3 \Rightarrow -4z + 17w = 0 \quad \text{--- (3)}$$

Put  $w = 4t$ ,  $t \in \mathbb{R}$

$$(3) \Rightarrow -4z + 17(4t) = 0$$

$$-4z = -17(4t)$$

$$\boxed{z = 17t}$$

$$(2) \Rightarrow y - 2(17t) + 2(4t) = 0$$

$$y - 34t + 8t = 0$$

$$\boxed{y = 26t}$$

$$(1) \Rightarrow x - 26t + 2(17t) - 3(4t) = 0$$

$$x - 26t + 34t - 12t = 0$$

$$x - 4t = 0$$

$$\boxed{x = 4t}$$

$$(x, y, z, w) = (4t, 26t, 17t, 4t)$$

$$\begin{aligned}
 11) (ii) \quad & x + 2y + z + 2w = 0 \\
 & 2x + 3y + 2z + 2w = 0 \\
 & 2x + 4y + 3z + 6w = 0 \\
 & 3x + 7y + 4z + 6w = 0
 \end{aligned}$$

$$x = -4t, y = 2t, z = -2t, w = t$$

2 MARKS

1) Find  $a$  and  $b$  if equations  
 $2x - 3y = 5$  &  $ax + by = -10$   
 have many solution

$$[A|B] = \left[ \begin{array}{cc|c} 2 & -3 & 5 \\ a & b & -10 \end{array} \right]$$

$$\sim \left[ \begin{array}{cc|c} 2 & -3 & 5 \\ 0 & 2b+3a & -20-5a \end{array} \right] \quad R_2 \rightarrow 2R_2 - aR_1$$

It is in echelon form,  
 given, the system has many solution,  
 $\Rightarrow R[A] = R[A|B] < n = 2$

$$\text{Hence } R[A] = R[A|B] = 1$$

$$\begin{aligned}
 \text{It is possible only } 2b + 3a &= 0 \quad \text{--- (1)} \\
 -20 - 5a &= 0 \quad \text{--- (2)}
 \end{aligned}$$

$$-5a = 20$$

$$\boxed{a = -4}$$

$$2b + 3(-4) = 0$$

$$2b = 12$$

$$\boxed{b = 6}$$

12) Find the condition satisfied by  $a, b, c$  so that the system of equation

$$x + 2y - 3z = a$$

$$3x - y + 2z = b$$

$$x - 5y + 8z = c$$

may have a solution.

$$[A|B] = \left[ \begin{array}{ccc|c} 1 & 2 & -3 & a \\ 3 & -1 & 2 & b \\ 1 & -5 & 8 & c \end{array} \right]$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & -7 & 11 & b-3a \\ 0 & -7 & 11 & c-a \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & -7 & 11 & b-3a \\ 0 & 0 & 0 & 2a-b+c \end{array} \right] R_3 \rightarrow R_3 - R_2$$

It is an Echelon Form

Given system have solution.

$$R(A) = R(A, B)$$

$$\text{Here } R(A) = 2 \Rightarrow R(A, B) = 2$$

It is possible only if

$$2a - b + c = 0$$

$$b = 2a + c$$

14) Write the algorithm for Gauss Jordan method

This method is a modification of Gauss Elimination method. In this method the coefficient matrix  $A$  of S.O.E  $\cancel{Ax=b}$   $Ax=B$  is to be diagonal matrix or unit matrix by making the matrix  $A$  not only upper triangular matrix but also LTM by making elements above the leading diagonal of  $A$  are also zeros 0's.

15) Write the diagonal dominance rule for a system of 3 linear Equations with 3 unknowns.

Consider S.O.E

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

Let us assume,

$$|a_1| > |b_1| + |c_1|$$

$$|b_2| > |a_2| + |c_2|$$

$$|c_3| > |a_3| + |b_3|$$

Diagonally Dominant :

S.O.E can be ordered so that each diagonal entry of the coefficient matrix is larger in magnitude than the sum of the magnitude of the other equation in that row. Such a system is called diagonally dominant.

16) Write Algorithm for Gauss Seidel method.

Consider SDE

$$a_1x + b_1y + c_1z = d_1 \text{ --- (1)}$$

$$a_2x + b_2y + c_2z = d_2 \text{ --- (2)}$$

$$a_3x + b_3y + c_3z = d_3 \text{ --- (3)}$$

Write the above eqns in DD. as

$$|a_1| > |b_1| + |c_1|$$

$$|b_2| > |a_2| + |c_2|$$

$$|c_3| > |a_3| + |b_3|$$

$$(1) \Rightarrow x = \frac{1}{a_1} [d_1 - b_1y - c_1z]$$

$$(2) \Rightarrow y = \frac{1}{b_2} [d_2 - a_2x - c_2z]$$

$$(3) \Rightarrow z = \frac{1}{c_3} [d_3 - a_3x - b_3y]$$

In general

$$x^{(r+1)} = \frac{1}{a_1} [d_1 - b_1y^{(r)} - c_1z^{(r)}]$$

$$y^{(r+1)} = \frac{1}{b_2} [d_2 - a_2x^{(r+1)} - c_2z^{(r)}]$$

$$z^{(r+1)} = \frac{1}{c_3} [d_3 - a_3x^{(r+1)} - b_3y^{(r+1)}] //$$

17) Solve SDE by Gauss Jordan Method.

$$(i) x_1 + 3x_2 + 3x_3 = 16$$

$$x_1 + 4x_2 + 3x_3 = 18$$

$$x_1 + 3x_2 + 4x_3 = 19$$

$$[A|B] = \left[ \begin{array}{ccc|c} 1 & 3 & 3 & 16 \\ 1 & 4 & 3 & 18 \\ 1 & 3 & 4 & 19 \end{array} \right]$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 3 & 3 & 16 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 0 & 3 & 10 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] R_1 \rightarrow R_1 - 3R_3$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] R_1 \rightarrow R_1 - 3R_3$$

It is an identity matrix

$$\Rightarrow x_1 = 1$$

$$x_2 = 2$$

$$x_3 = 3$$

$$(ii) \begin{aligned} x + y + z &= 9 \\ 2x - 3y + 4z &= 13 \\ 3x + 4y + 5z &= 40 \end{aligned}$$

$$[A|B] = \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & -3 & 4 & 13 \\ 3 & 4 & 5 & 40 \end{array} \right]$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -5 & 2 & -5 \\ 0 & 1 & 2 & 13 \end{array} \right] \begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \end{aligned}$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -5 & 2 & -5 \\ 0 & 0 & 12 & 125 \end{array} \right] R_3 \rightarrow 5R_3 + R_2$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -5 & 2 & -5 \\ 0 & 0 & 1 & 5 \end{array} \right] R_3 \rightarrow \frac{R_3}{12}$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & -5 & 0 & -15 \\ 0 & 0 & 1 & 5 \end{array} \right] \begin{aligned} R_1 &\rightarrow R_1 - R_3 \\ R_2 &\rightarrow R_2 - 2R_3 \end{aligned}$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 5 \end{array} \right] R_2 \rightarrow \frac{R_2}{-5}$$

$$(v) \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 5 \end{array} \right] \quad R_1 \rightarrow R_1 - R_2$$

$$x_1 = 1$$

$$x_2 = 3$$

$$x_3 = 5$$

$$(iii) \quad 2x_1 + x_2 + x_3 = 10$$

$$3x_1 + 2x_2 + 3x_3 = 18$$

$$x_1 + 4x_2 + 9x_3 = 16$$

$$[A|B] = \left[ \begin{array}{ccc|c} 2 & 1 & 1 & 10 \\ 3 & 2 & 3 & 18 \\ 1 & 4 & 9 & 16 \end{array} \right]$$

$$(v) \left[ \begin{array}{ccc|c} 1 & 4 & 9 & 16 \\ 3 & 2 & 3 & 18 \\ 2 & 1 & 1 & 10 \end{array} \right] \quad R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

$$x_1 = 1$$

$$x_2 = 3$$

$$x_3 = 2$$

$$01 = 8x_1 + 5x_2 + x_3 = 10$$

$$81 = 8x_1 + 9x_2 + 3x_3 = 18$$

$$11 = 8x_1 + 4x_2 + x_3 = 16$$

$$\begin{bmatrix} 8 & 5 & 1 \\ 8 & 9 & 3 \\ 8 & 4 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 18 \\ 16 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 1 \\ 8 & 9 & 3 \\ 8 & 1 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 18 \\ 16 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 4 & 1 \\ 8 & 9 & 3 \\ 8 & 1 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 18 \\ 16 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 18 \\ 16 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 18 \\ 0 \end{bmatrix}$$

18. Solve SDE by Gauss Seidel method <sup>correct to</sup> ~~collective~~ four decimal places.

$$\begin{aligned} \text{(i)} \quad & 9x_1 + 2x_2 + 4x_3 = 10 \\ & 2x_1 - 4x_2 + 10x_3 = -15 \\ & x_1 + 10x_2 + 4x_3 = 6 \end{aligned}$$

$$[A|B] = \left[ \begin{array}{ccc|c} 9 & 2 & 4 & 10 \\ 2 & -4 & 10 & -15 \\ 1 & 10 & 4 & 6 \end{array} \right]$$

The above eqns can be written via dd method.

$$9x_1 + 2x_2 + 4x_3 = 10 \quad \text{---(1)}$$

$$x_1 + 10x_2 + 4x_3 = 6 \quad \text{---(2)}$$

$$2x_1 - 4x_2 + 10x_3 = -15 \quad \text{---(3)}$$

$$(1) \Rightarrow x_1 = \frac{1}{9} [10 - 2x_2 - 4x_3]$$

$$(2) \Rightarrow x_2 = \frac{1}{10} [6 - x_1 - 4x_3]$$

$$(3) \Rightarrow \frac{1}{10} x_3 = \frac{1}{10} [-15 - 2x_1 + 4x_2]$$

Let  $x_2 = x_3 = 0$  (initial values) assume.

I. Iteration:

$$x_1^{(1)} = \frac{1}{9} [10] = 1.1111$$

$$x_2^{(1)} = \frac{1}{10} [6 - 1.1111 - 4(0)] = 0.4889$$

$$x_3^{(1)} = \frac{1}{10} [-15 - 2(1.1111) + 4(0.4889)]$$

$$x_3^{(1)} = -1.5267$$

II. Iteration:

$$x_1^{(2)} = \frac{1}{9} [10 - 2(0.4889) - 4(1.5267)]$$
$$= 1.6810$$

$$x_2^{(2)} = \frac{1}{10} [6 - 1.6810 - 4(-1.5267)] = 1.0426$$

$$x_3^{(2)} = \frac{1}{10} [-15 - 2(1.6810) + 4(1.0426)]$$
$$= -1.4192$$

III. Iteration:

$$x_1^{(3)} = \frac{1}{9} [10 -$$